1.4 Modeling with Linear Functions

Name _____ Period ____ Date _____

Algebra II 1.4 · Student sheet

I can…

- write equations of linear functions from graphs and tables.
- find a line of fit and the line of best fit.
- interpret the correlation coefficient and use linear models.

Vocabulary · fill in the missing word

Slope-intercept form — $y =$ _____

Point-slope form — $y - y_1 =$ _____

Line of fit — a line drawn close to most of the points of a _____ plot

Line of best fit — the line that fits the data most _____ (from technology)

Correlation coefficient r — a number from −1 to 1 that _____ how well a line fits; near ±1 is strong, near 0 is weak

Part 1 · Study time

Hours studied	1	2	3	4	5	6	7	8
Score	62	68	71	75	80	83	88	90

Plot the data, draw a line of fit, and choose two points ON your line: (_____) and (_____)

Your equation: _____ Technology: y = _____ r = _____

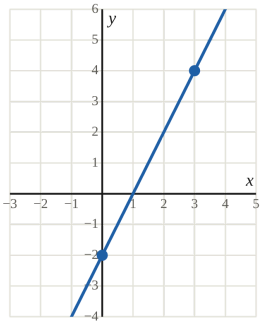
What does the slope mean? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Equations of lines: Slope m = _____; $y = mx + b$; $y - y_1 =$ _____

Example 1

Write an equation of the line.



Example 2

The table shows the total cost of a gym membership. Write an equation for the cost y after x months. What do the slope and y-intercept mean?

Months, x	0	2	4	6
Cost, y ($)	40	100	160	220

Line of fit: A line of fit is drawn close to _____ of the points.

Example 3

Use technology to find the line of best fit for the study-time data and the correlation coefficient. Predict the score for 10 hours of studying.

Hours studied	1	2	3	4	5	6	7	8
Score	62	68	71	75	80	83	88	90

Correlation coefficient: r near 1: _____. r near −1: _____. r near 0: _____.

Example 4

Describe the correlation shown by each correlation coefficient.

a. $r = -0.94$ b. $r = 0.12$ c. $r = 0.85$

Part 3 · Partner practice

Partner 1

Write an equation of the line through (−1, 7) and (3, −1).

Partner 2

Write an equation of the line through (6, 1) that is perpendicular to $y = 3x - 2$.